Nonlinear Unmixing for Hyperspectral Images via Kernel-Transformed Bilinear Mixing Models

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***Abstract*— Due to the presence of multiple scatterings, linear unmixing methods may not perform well in practical applications, and thus nonlinear unmixing has become an urgent problem to be solved. Usually, the mixing process in the observed scenarios is physically based, and many well-designed models have been proposed to interpret it. Recently, kernel-based nonlinear unmixing methods have been popularly studied to achieve a model-free and flexible representation of the nonlinearity. However, the existing kernel-based methods are mainly data-driven, which could make them fail to match the real physical mixing mechanism and result in the occurrence of overfitting. In this article, a kernel-based bilinear unmixing (KBU) method was proposed to transform the classic bilinear mixing models into their equivalent kernel forms that are more general and effective in expressing second-order scatterings. Two specific types of kernel transformations were designed, and the alternating direction method of multipliers (ADMM) was used to solve the kernel-transformed model-based optimization problem for unmixing. Moreover, the spatial prior was exploited to further improve the unmixing accuracy, and here we employ the total variation (TV) regularization as a paradigm. Experiments on synthetic data sets, physics-based simulated data sets, and real data were conducted to evaluate the algorithms. It is validated that our methods have better performance in abundance estimation and nonlinear reconstruction compared with other nonlinear unmixing methods.**

***Index Terms*— Bilinear mixing models (BMMs), hyperspectral images (HSIs), kernel transformation, nonlinear unmixing, total variation (TV) regularization.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain a wealth of spatial distribution as well as detailed spectral information of ground objects for hundreds of narrow spectral bands from ultraviolet and visible to shortwave infrared for every pixel [1]. Over the decades, hyperspectral remote sensing imagery has been widely used in various applications, including agriculture, environment monitoring, and natural resources surveying [2], [3]. However, due to the limitation of the spatial resolution of HSIs, multiple ground objects may exist in the same area and be observed in one pixel, generating the so-called mixed pixel. Generally, mixed pixels often widespread in HSIs and significantly influence the performance of traditional pixel-level applications [4]. To get rid of the above dilemma, spectral unmixing (SU) is usually used as an effective tool to decompose the mixed pixels into a collection of endmembers and their corresponding fractional abundances [3]. In this way, applications can be executed at the subpixel level with better performance.

Depending on the physical mixing processes of distinct substance spectra, their corresponding mixing models can be derived mathematically, thus facilitating the implementation of SU. The linear mixing model (LMM) is the most popular theoretical model that has been successfully used for unmixing. It simply assumes that light interacts with one material and then reaches the sensors directly, so each pixel in the HSIs can be simply represented as a linear combination of endmembers [2]. Based on the above assumption, this model performs effectively in macroscopic mixtures. Popular algorithms including N-finder (N-FINDR) [5], orthogonal bases algorithm (OBA) [6], vertex component analysis (VCA) [7], and simplex growing algorithm (SGA) [8] serve for linear endmember extraction. Fully constrained least squares (FCLS) [9] aims at abundance inversion using prior endmembers. Although linear unmixing has been recognized in a large number of applications, the LMM may fail to provide good unmixing results when the nonlinear mixing effects are nonnegligible in some scenarios [10], [11]. The nonlinear mixing effects mainly come from the multiple scatterings among different substances and are enhanced especially when the mixing scale is microscopic [12], [13]. For example, intimate mixtures come into being when complex multiple scatterings happen among microscopic mineral particles [14]; multilayered mixtures result from the light's reflections among substances in different layers, common in vegetated and urban areas [15]. Different from linear SU, there is still a considerable challenge besides great significance in the study of nonlinear SU due to its complexity.

In spite of the intractability of the nonlinear mixing effects, many nonlinear mixing models (NMMs) and their corresponding unmixing algorithms have been developed so far [10]. The famous Hapke model [16] was proposed to unmix intimate mixtures based on the radiative transfer equations. However, this model requires not few prior scenes-dependent physical parameters, which are difficult to be acquired in reality. As a result, the practical application of such a model is a bit limited. To avoid the dependence of large numbers of physical parameters, many NMMs are developed by incorporating nonlinear mixing terms into the LMM [10]. The bilinear mixing models (BMMs) extend the LMM by considering the second-order scatterings and constructing the nonlinear mixing

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terms by the Hadamard products of every two endmembers and their corresponding abundances [11]. The common BMMs include the Nascimento model (NM) [12], Fan model (FM) [17], generalized bilinear model (GBM) [18], and polynomial post-nonlinear model (PPNM) [19]. Given the simplicity of mathematical expression of BMMs, many BMM-based unmixing algorithms have been proposed, such as the Bayesian approaches, gradient descent algorithm (GDA) [18], [19], nonnegative matrix factorization (NMF) [20], [21], nonnegative tensor factorization (NTF) [22], [23], and some heuristic approaches [24]–[26]. The second-order scatterings are often dominant in the nonlinear mixing effects, and the higher-order interactions contribute little. This fact facilitates the superior performance of the BMM-based unmixing methods in most real cases [10]. In addition, some spectral mixing models including high-order scatterings were recently proposed, such as the multilinear mixing model (MLM) [27]. However, the computational cost of these models and algorithms is an intractable problem.

The above nonlinear unmixing algorithms assume the mixing process is a specific known prior [10], [11]. But the actual nonlinear mixing mechanisms may be very complex, leading that valid prior knowledge is rare in real cases [28]. Accordingly, model-free nonlinear unmixing methods come into consideration which seek flexible representation from the data. The popular model-free strategies include manifold-based approaches [29], [30], and kernel-based algorithms [31]–[34]. The kernel-based methods have shown great potential in interpreting the nonlinearity resulting from single scatterings to infinite order scatterings in the reproducing kernel Hilbert spaces (RKHS) [35]. Their efficiency has been proved in addressing nonlinear problems. A class of kernel-based unmixing methods have already been investigated, such as the kernel versions of principal component analysis (PCA) [36], FCLS [33], and NMF [31]. These algorithms were obtained by replacing the inner products between the endmember spectra from their original methods with kernel functions. Essentially, this is a high-dimensional mapping of the endmember spectra, independent of the interactions between the endmembers, but lacking in physical interest. Unlike the traditional use of kernel functions, K-Hype [32] takes nonlinear interactions of the endmembers into account. Specifically, a pixel is redefined as the combination of a linear term and a nonlinear fluctuation term, which takes advantage of parameter-free representations in the RKHS. Therefore, arbitrary nonlinear mixtures can be conveniently modeled [32]. This flexibility allows it to have superior performance in practical tasks.

However, K-Hype reconstructs the nonlinearity in a data-driven way and attaches little importance to the mixing mechanisms of the nonlinearity, leading to doubts about what the reconstructed nonlinear part in a pixel contains after the kernel-based model-free representation. Gaussian kernel and polynomial kernel functions are commonly used owing to their powerful representation abilities [37]. Although they can map vectors to higher dimensions in the RKHS, the mapped vectors contain not only second-order components, but also high-order components (especially for the Gaussian kernel which includes infinite-order components), zero-order, and first-order components. The zero-order and high-order components tend to over-fit the results when the nonlinear part in a pixel does not require these components. Moreover, the first-order components overlap the linear part in the pixel, so this model-free representation will interfere with the reconstruction of the linear part in unmixing. Hence, if the mixing mechanism can be highlighted and the physical-based NMMs can be combined, the kernel-based approaches may better interpret the nonlinearity and achieve higher accuracy of abundance estimation.

On the other hand, most existing nonlinear unmixing algorithms focus on individual pixels without considering the spatial relationships. This drawback may come from the fact that nonlinear unmixing itself is a highly complex task [38]. However, it is a waste to ignore this powerful prior during the nonlinear unmixing process, which, after all, can effectively improve the accuracy of unmixing, which has been validated in many state-of-the-art linear unmixing algorithms [39]–[41]. Usually, the introduction of spatial regularizers increases the algorithm's complexity, especially for nonlinear unmixing algorithms, which makes them more challenging to be solved. Therefore, it is essential to ensure low unmixing complexity while introducing spatial regularization terms and construct a compact and reliable constrained optimization problem for nonlinear unmixing.

In this article, we present a kernel-based bilinear unmixing (KBU) method. First, we extend the LMM by introducing a kernel-based bilinear term. In this section, two kernels are specifically designed to cover the second-order scattering components, mainly based on the assumption that the second-order interactions dominate the nonlinear mixing effects [10]. Unlike the conventional Gaussian or polynomial kernels, they nicely interpret the nonlinear part without interference and essentially decrease the overfitting results induced by higher-order components. Kernel-based representations give a nonparametric expression of the function space and can achieve a more flexible interpretation. Then, we construct a concise constrained optimization problem for KBU and solve it using alternating direction methods of multipliers (ADMM) [39]. One remarkable advantage of our approach is that the numerical fitting of the nonlinear part only relies on the kernel-based representations where the direct participation of the abundances can be omitted. Therefore, during the iterative process, the optimization subproblem for the abundances remains as a linear unmixing problem, which can significantly reduce the algorithm's complexity and facilitates future generalizations to the method. Moreover, owing to this optimization problem's conciseness, the spatial prior could be easily embedded into it. Specifically, the total variation (TV) regularization [40] is adopted as a paradigm to further improve the unmixing accuracy.

The main contributions of this article can be briefly summarized as follows.

1) A KBU method is proposed. Two specially designed kernels exploit the mixing mechanism and efficiently transform the popular BMMs to better fit the nonlinearity. Remarkably, the flexible kernel-based nonparametric representations for the nonlinear part require no direct

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participation of the abundances, which significantly eases the optimization.

2) The spatial prior is conveniently incorporated into the proposed methods without burdening the optimization process, and specifically, the TV regularizer is utilized as a paradigm. The ADMM-driven constrained optimization for the KBU problem is constructed to achieve accurate unmixing results efficiently.

The remainder of this article is organized as follows. Section II reviews the BMM and K-Hype. In Section III, the proposed KBU is first described in detail. Then, the corresponding constrained optimization problem is constructed, which is further reformulated by using the spatial TV regularization term, and the unmixing algorithm is deduced. In Section IV, experiments are conducted on model-based synthetic data, physical-based simulated data, and real hyperspectral data. Finally, the conclusions are drawn in Section V.

II. Related Works

A. BMMs

Denoting $\boldsymbol{X} \in \mathbb{R}^{L\times N}$ an HSI with N pixels and L spectral bands. In the LMM, each pixel $\boldsymbol{x}_j \in \mathbb{R}^{L\times 1}(j = 1,\ldots,N)$ is denoted as a linear combination of endmembers and their corresponding abundances [43]. Besides, abundances are subject to two constraints, i.e., abundances' nonnegativity constraint (ANC) and abundances' sum-to-one constraint (ASC) [9]. Generally, the LMM has the following mathematical expression:

$$\begin{aligned}\boldsymbol{x}_j &= \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + \boldsymbol{e}_j \\ \text{s.t. } a_{i,j} &\geq 0, \quad \sum_{i=1}^{R} a_{i,j} = 1\end{aligned} \tag{1}$$

where $\boldsymbol{m}_i \in \mathbb{R}^{L\times 1}$ represents the ith endmember in the endmember matrix $\boldsymbol{M} \in \mathbb{R}^{L\times R}$, R is the number of the endmembers, and $a_{i,j}$ is its corresponding abundance in $\boldsymbol{x}_j$.

Based on the LMM, we review three typical BMMs, i.e., FM [17], GBM [18], and PPNM [19]. They all consist of a linear part as well as a nonlinear part made up of bilinear terms. For the FM, it follows the assumption that the intensity of the second-order scatterings in the nonlinear part is equal to the product of the abundances of the corresponding endmembers. Each pixel has the following mathematical expression:

$$\boldsymbol{x}_j = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + \sum_{i=1}^{R-1}\sum_{k=i+1}^{R} (\boldsymbol{m}_i \odot \boldsymbol{m}_k) a_{i,j} a_{k,j} + \boldsymbol{e}_j \tag{2}$$

where $\odot$ denotes the Hadamard product, and $\boldsymbol{m}_i \odot \boldsymbol{m}_k$ is a virtual endmember, which stands for the second-order interactions between endmembers $\boldsymbol{m}_i$ and $\boldsymbol{m}_k$.

The GBM has a more flexible form compared to the FM, and it can be regarded as a generalization of the FM with extra parameters

$$\boldsymbol{x}_j = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + \sum_{i=1}^{R-1}\sum_{k=i+1}^{R} \gamma_{i,j,k} (\boldsymbol{m}_i \odot \boldsymbol{m}_k) a_{i,j} a_{k,j} + \boldsymbol{e}_j \tag{3}$$

TABLE I
Parameters of Three BMMs

Model	t_1	t_2	$\zeta_{i,j,k}$
FM	$R-1$	$i+1$	1
GBM	$R-1$	$i+1$	$\gamma_{i,j,k} \in [0,1]$
PPNM	R	1	$b_j \in \mathbb{R}$

where $\gamma_{i,j,k} \in [0, 1]$ is a nonlinear parameter to adjust the intensity of the nonlinear effects. The flexibility of the GBM is demonstrated by the fact that the GBM becomes the FM when $\gamma_{i,j,k} = 1, \forall i, j, k$ and degrades to the LMM when $\gamma_{i,j,k} = 0, \forall i, j, k$. Therefore, it is more suitable for practical tasks than the FM.

The PPNM further studies the bilinear interaction generated from the same endmembers, and it can be given as follows:

$$\boldsymbol{x}_j = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + b_j \sum_{i=1}^{R}\sum_{k=1}^{R} (\boldsymbol{m}_i \odot \mathbf{m}_k) a_{i,j} a_{k,j} + \boldsymbol{e}_j \tag{4}$$

where $b_j \in \mathbb{R}$ measures the contribution of the nonlinear interactions in each pixel. The PPNM has a certain degree of generalization like the GBM, such that it can reduce to the LMM when b_j is set as 0. In addition, the PPNM shows less complexity than the GBM because only one nonlinear parameter b_j is required to regulate the nonlinear part for each pixel.

The above three BMMs have similar mathematical expressions, and thus a general form can be derived as follows:

$$\boldsymbol{x}_j = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + \sum_{i=1}^{t_1}\sum_{k=t_2}^{R} \zeta_{i,j,k} (\boldsymbol{m}_i \odot \boldsymbol{m}_k) a_{i,j} a_{k,j} + \boldsymbol{e}_j \tag{5}$$

where $\zeta_{i,j,k}$ denotes the nonlinear parameter of each BMM and its detailed definition is summarized in Table I.

B. K-Hype

In this section, we review the kernel-based unmixing method introduced in [32]. As discussed in [32], each pixel $\boldsymbol{x}_j \in \mathbb{R}^{L\times 1}$ $(j = 1,\ldots,N)$ in an HSI can be modeled as follows:

$$x_{j,l} = \boldsymbol{\psi}_{\boldsymbol{a}_j}(\tilde{\boldsymbol{m}}_l) + e_{j,l}, \quad l = 1,\ldots,L \tag{6}$$

where $x_{j,l}$ is the lth band of the pixel $\boldsymbol{x}_j$, $\tilde{\boldsymbol{m}}_l \in \mathbb{R}^{1\times R}$ is the lth row of the endmember matrix $\boldsymbol{M} \in \mathbb{R}^{L\times R}$, and $e_{j,l}$ is denoted as the error of the lth band. $\boldsymbol{\psi}_{\boldsymbol{a}_j}$ is an unknown nonlinear function that defines the interactions between the endmember spectra parameterized by their fractional abundances $\boldsymbol{a}_j \in \mathbb{R}^{R\times 1}$ [32]. It is rational that this model is flexible enough to construct most types of complex light interactions occurring in the HSIs. The key is to find a suitable function $\boldsymbol{\psi}_{\boldsymbol{a}_j}$ that can describe the mixing process accurately. The kernel-based smooth function representation is a powerful tool to learn it directly from the data [37], [43]. Thus, a general problem is formulated as

$$\boldsymbol{\psi}^*_{\boldsymbol{a}_j} = \arg\min_{\boldsymbol{\psi}_{\boldsymbol{a}_j}} \frac{1}{2}\left\|\boldsymbol{\psi}_{\boldsymbol{a}_j}\right\|^2_{\boldsymbol{\mathcal{H}}} + \frac{1}{2\mu}\sum_{l=1}^{L}\left(x_{j,l} - \boldsymbol{\psi}_{\boldsymbol{a}_j}(\tilde{\boldsymbol{m}}_l)\right)^2 \tag{7}$$

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where $\mathcal{H}$ is an RKHS [35], and μ is a positive parameter that balances the reconstruction error (RE) and the regularity of the function $\boldsymbol{\psi}_{\boldsymbol{a}_j}$. It is clear that the function $\boldsymbol{\psi}_{\boldsymbol{a}_j}$ of $\mathcal{H}$ may not be adequately and finitely parameterized and lacks physical meanings. Therefore, a linear trend parameterized by the abundance vector $\boldsymbol{a}_j$ combined with a nonlinear fluctuation function $\boldsymbol{\psi}_j$ is designed [32]

$$\boldsymbol{\psi}_{\boldsymbol{a}_j}(\tilde{\boldsymbol{m}}_l) = \tilde{\boldsymbol{m}}_l \boldsymbol{a}_j + \boldsymbol{\psi}_j(\tilde{\boldsymbol{m}}_l) \tag{8}$$

where $\boldsymbol{\psi}_j : \mathbb{R}^R \to \mathbb{R}$ is an arbitrary real-valued function of an RKHS $\mathcal{H}$, endowed with the reproducing kernel κ such that $\boldsymbol{\psi}_j(\tilde{\boldsymbol{m}}_l) = \langle \boldsymbol{\psi}_j, \kappa(\cdot, \tilde{\boldsymbol{m}}_l)\rangle_{\mathcal{H}}$. As a result, the linear part is separated out in an explicit way, and only the nonlinear part needs the estimation of the function $\boldsymbol{\psi}_j$. In [32], a least square support vector regression (LS-SVR) problem [44] is introduced to solve the nonlinear unmixing problem based on (7) and (8)

$$\begin{aligned} &\boldsymbol{a}_j^*, \boldsymbol{\psi}_j^* \\ &= \underset{a_j,\ \psi_j}{\arg\min} \frac{1}{2}\left(\|\boldsymbol{a}_j\|^2 + \|\boldsymbol{\psi}_j\|_{\mathcal{H}}^2 + \frac{1}{\mu}\|\boldsymbol{e}_j\|^2\right) \\ &\quad \text{s.t. } \boldsymbol{a}_j \geq 0, 1^{\mathrm{T}}\boldsymbol{a}_j = 1, e_{j,l} = x_{j,l} - \tilde{\boldsymbol{m}}_l \boldsymbol{a}_j - \boldsymbol{\psi}_j(\tilde{\boldsymbol{m}}_l), \\ &\quad l = 1, \ldots, L \end{aligned} \tag{9}$$

where $\boldsymbol{a}_j^*$ and $\boldsymbol{\psi}_j^*$ are the estimated abundance vector and nonlinear function for the jth pixel, respectively. The constrained optimization problem is convex, and it can be solved using the standard duality theory [44].

III. Proposed Method

A. Kernel-Transformed BMM

The kernel-based approach K-Hype shows its great flexibility and potential in explaining nonlinear mixing pixels. The combination of linear trends and nonlinear fluctuation terms makes the data-driven fitting of nonlinear functions occur mainly in the nonlinear part. However, the mixing mechanism in the pixels is physical-based, and the purely data-driven kernel approach will inevitably result in overfitting, which severely affects the unmixing accuracy. Therefore, for nonlinear hyperspectral unmixing, the mixing mechanism should be taken into account to better adapt the kernel-based approach to this problem.

Considering the dominance of second-order scatterings in the nonlinear mixing effects and the wide application of the BMMs, in this article, we propose a KBU method named KBU. Specifically, for the proposed algorithm, the nonlinear part only includes the second-order components. The estimation of the nonlinear part does not produce a negative interference with the linear part as compared to other kernel-based NMMs [29]–[32]. In this way, it overcomes the defects in the model-free approaches. Moreover, the kernel-based representation still maintains the advantage of a nonparametric representation, providing a more flexible pixel interpretation than the popular BMMs [17]–[19]. To be specific, a single pixel can be expressed as

$$\boldsymbol{x}_j = \boldsymbol{M}\boldsymbol{a}_j + \boldsymbol{K}\boldsymbol{\beta}_j \tag{10}$$

where $\boldsymbol{K} \in \mathbb{R}^{L\times L}$ is a Gram matrix for kernel transformation defined as $[\boldsymbol{K}]_{ij} = \kappa(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j)$, and $\boldsymbol{\beta}_j \in \mathbb{R}^{L\times 1}$ denotes the nonlinear parameter vector with respect to the pixel $\boldsymbol{x}_j$. For (10), the nonlinear components in the pixel are represented by the second term, only for the bilinear components. $\boldsymbol{\beta}_j$ is used to adjust the bilinear components.

B. Design of the Kernel and Its Rationality

The Gaussian kernel and polynomial kernel have been widely used in the literature due to their powerful interpretation capability [31]–[34]. They commonly have the following expressions:

$$\kappa(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j) = \exp\left(-\frac{\|\tilde{\boldsymbol{m}}_i - \tilde{\boldsymbol{m}}_j\|^2}{2\sigma^2}\right) \tag{11}$$

$$\kappa(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j) = \left(\tilde{\boldsymbol{m}}_i \tilde{\boldsymbol{m}}_j^{\mathrm{T}} + c\right)^d. \tag{12}$$

Equations (11) and (12) actually correspond to an inner product in the RKHS

$$\kappa\left(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j\right) = \phi(\tilde{\boldsymbol{m}}_i)\phi\left(\tilde{\boldsymbol{m}}_j\right)^{\mathrm{T}}. \tag{13}$$

For instance, given an endmember matrix $\boldsymbol{M} \in \mathbb{R}^{L\times R}$ and $d = 2, c = 1$, in the case of the polynomial kernel, a possible $\phi(\tilde{\boldsymbol{m}}_i)$ can be expressed as

$$\begin{aligned} \phi(\tilde{\boldsymbol{m}}_i) = \Big(&1, \sqrt{2}\tilde{\boldsymbol{m}}_i, m_{i,1}^2, \\ &\ldots, m_{i,R}^2, \sqrt{2}\, m_{i,1}m_{i,2}, \ldots, \sqrt{2}\, m_{i,R-1}m_{i,R}\Big). \end{aligned} \tag{14}$$

In terms of (8) in K-Hype, although the function $\boldsymbol{\psi}_j$ to be estimated only works for the nonlinear part, zero-order and first-order terms in (14) could severely interfere with the abundance estimation in the linear part. Moreover, for the Gaussian kernel, the infinite-order components inevitably result in the overfitting of the nonlinear part, indicating that these kernels may not be theoretically suitable for the nonlinear unmixing tasks.

Since the nonlinear part in (10) requires only the consideration of the second-order scatterings, the selected kernels should cover all possible second-order scattering components. If bilinear interactions from the same endmembers are not considered like the FM and the GBM, an available kernel κ_{GBM} and its corresponding Gram matrix $\boldsymbol{K}_{\mathrm{GBM}}$ can be given as follows:

$$\begin{cases} \kappa_{\mathrm{GBM}}(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j) = [\boldsymbol{K}_{\mathrm{GBM}}]_{i,j} = \left(\tilde{\boldsymbol{m}}_i \tilde{\boldsymbol{m}}_j^{\mathrm{T}}\right)^2 - \tilde{\boldsymbol{m}}_i^2 \tilde{\boldsymbol{m}}_j^{2T} \\ \boldsymbol{K}_{\mathrm{GBM}} = (\boldsymbol{M}\boldsymbol{M}^{\mathrm{T}})^2 - \boldsymbol{M}^{2T}\boldsymbol{M}^2 \end{cases} \tag{15}$$

where $(\cdot)^2$ denotes the squaring operation on each element of the matrix.

For the above kernel, a possible $\phi(\tilde{\boldsymbol{m}}_i)$ in (13) can be denoted as

$$\phi(\tilde{\boldsymbol{m}}_i) = \sqrt{2}\left(m_{i,1}m_{i,2},\ m_{i,1}m_{i,3},\ \ldots,\ m_{i,R-1}m_{i,R}\right). \tag{16}$$

It is clear that the high-dimensional mapped vectors include all the possible bilinear components. In this sense, the nonlinear term on the lth band can be written as

$$(\boldsymbol{K}_{\mathrm{GBM}}\boldsymbol{\beta})_l = \boldsymbol{\phi}(\tilde{\boldsymbol{m}}_l)\left(\sum_{i=1}^{L}\boldsymbol{\phi}(\tilde{\boldsymbol{m}}_i)\beta_i\right)^{\mathrm{T}} = \sum_{i=1}^{R-1}\sum_{j=i+1}^{R} b_{i,j}m_{l,i}m_{l,j} \tag{17}$$

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where $b_{i,j}(i = 1, \ldots, R-1; j = i+1, \ldots, R)$ denotes the bilinear parameters to be estimated. Generally, this formula can cover all the cases in the FM and GBM, and their specific relationships can be demonstrated as follows.

For a pixel $\boldsymbol{x}_j$ based on the GBM, if we define a specific form of the nonlinear parameter $\zeta_{i,k,j}$ in (18), the relationship between the GBM and its kernel-transformed form in (15) can be derived in the following equation:

$$\zeta_{i,k,j} = \frac{2\sum_{l=1}^{L}\beta_{l,j}m_{l,i}m_{l,k}}{a_{i,j}a_{k,j}} \tag{18}$$

$$\begin{aligned}\boldsymbol{x}_j &= \sum_{i=1}^{R}\boldsymbol{m}_i a_{i,j} + \sum_{i=1}^{R-1}\sum_{k=i+1}^{R}\zeta_{i,k,j}(\boldsymbol{m}_i \odot \boldsymbol{m}_k)a_{i,j}a_{k,j} \\ &= \sum_{i=1}^{R}\boldsymbol{m}_i a_{i,j} + \sum_{i=1}^{R-1}\sum_{k=i+1}^{R}\left(2\sum_{l=1}^{L}\beta_{l,j}m_{l,i}m_{l,k}\right)(\boldsymbol{m}_i \odot \boldsymbol{m}_k) \\ &= \sum_{i=1}^{R}\boldsymbol{m}_i a_{i,j} + \begin{bmatrix} \phi(\tilde{\boldsymbol{m}}_1)\left(\sum_{i=1}^{L}\phi(\tilde{\boldsymbol{m}}_i)\beta_{i,j}\right)^{\mathrm{T}} \\ \phi(\tilde{\boldsymbol{m}}_2)\left(\sum_{i=1}^{L}\phi(\tilde{\boldsymbol{m}}_i)\beta_{i,j}\right)^{\mathrm{T}} \\ \vdots \\ \phi(\tilde{\boldsymbol{m}}_L)\left(\sum_{i=1}^{L}\phi(\tilde{\boldsymbol{m}}_i)\beta_{i,j}\right)^{\mathrm{T}} \end{bmatrix} \\ &= \boldsymbol{M}\boldsymbol{a}_j + \begin{bmatrix} (\boldsymbol{K}_{\mathrm{GBM}}\boldsymbol{\beta}_j)_1 \\ (\boldsymbol{K}_{\mathrm{GBM}}\boldsymbol{\beta}_j)_2 \\ \vdots \\ (\boldsymbol{K}_{\mathrm{GBM}}\boldsymbol{\beta}_j)_L \end{bmatrix} \\ &= \boldsymbol{M}\boldsymbol{a}_j + \boldsymbol{K}_{\mathrm{GBM}}\boldsymbol{\beta}_j. \end{aligned} \tag{19}$$

It can be validated that the GBM and our model are derived equivalently, indicating the rationality of our model. With this design, our model actually relaxes the GBM, and can give a more flexible representation of the secondary scattering by means of kernel tricks to facilitate subsequent calculations.

On the other hand, when bilinear interactions from the same endmembers are considered like the PPNM, a kernel κ_{PPNM} and its Gram matrix $\boldsymbol{K}_{\mathrm{PPNM}}$ can be written as

$$\begin{aligned}\kappa_{\mathrm{PPNM}}(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j) &= [\boldsymbol{K}_{\mathrm{PPNM}}]_{i,j} = \left(\tilde{\boldsymbol{m}}_i\tilde{\boldsymbol{m}}_j^{\mathrm{T}}\right)^2 \\ \boldsymbol{K}_{\mathrm{PPNM}} &= (\boldsymbol{M}\boldsymbol{M}^{\mathrm{T}})^2 \end{aligned} \tag{20}$$

and the similar formulas in (16) and (17) can be derived

$$\begin{aligned}\phi(\tilde{\boldsymbol{m}}_i) = \Big(&m_{i,1}m_{i,1},\ \ldots,\ m_{i,R}m_{i,R},\ \sqrt{2}\,m_{i,1}m_{i,2}, \\ &\ldots,\ \sqrt{2}\,m_{i,R-1}m_{i,R}\Big)\end{aligned} \tag{21}$$

$$(\boldsymbol{K}_{\mathrm{PPNM}}\boldsymbol{\beta})_l = \boldsymbol{\phi}(\tilde{\boldsymbol{m}}_l)\left(\sum_{i=1}^{L}\boldsymbol{\phi}(\tilde{\boldsymbol{m}}_i)\beta_i\right)^{\mathrm{T}} = \sum_{i=1}^{R}\sum_{j=i}^{R}b_{i,j}m_{l,i}m_{l,j}. \tag{22}$$

Compared with κ_{GBM}, κ_{PPNM} supplements the bilinear terms of the same endmembers, which is consistent with the case described in the PPNM. Their intimate connection can be explicitly revealed as what has been mentioned above.

Obviously, both of the above Gram matrices are semipositive definite and follow the Mercer's representation theorem [45]. Different from the conventional Gaussian kernel and polynomial kernel, the designed kernels κ_{GBM} and κ_{PPNM} are more reasonable for fitting the nonlinear parts.

C. Problem Formulation and the Unmixing Algorithm

For a single pixel $\boldsymbol{x}_j$, the abundance vector $\boldsymbol{a}_j$ and the nonlinear parameter vector $\boldsymbol{\beta}_j$ should be estimated, and thus, we realize hyperspectral nonlinear unmixing by solving the following constrained optimization problem:

$$\begin{aligned}\boldsymbol{a}_j^*, \boldsymbol{\beta}_j^* = \underset{\boldsymbol{\alpha}_j, \boldsymbol{\beta}_j, \boldsymbol{e}_j}{\arg\min}\ &\frac{1}{2}\|\boldsymbol{e}_j\|^2 + \frac{\gamma}{2}\boldsymbol{\beta}_j^{\mathrm{T}}\boldsymbol{K}\boldsymbol{\beta}_j \\ &\text{s.t. } \mathbf{1}^{\mathrm{T}}\boldsymbol{a}_j = 1, \boldsymbol{a}_j \geq 0, \boldsymbol{x}_j = \boldsymbol{M}\boldsymbol{a}_j + \boldsymbol{K}\boldsymbol{\beta}_j + \boldsymbol{e}_j\end{aligned} \tag{23}$$

where γ is a positive parameter that controls the tradeoff between the model fitting and the regularity of the nonlinear parameter, which limits the overfitting of $b_{i,j}$ in (17). $\boldsymbol{e}_i$ denotes the error in the pixel. In this article, $\boldsymbol{K}$ will be selected as $\boldsymbol{K}_{\mathrm{GBM}}$ or $\boldsymbol{K}_{\mathrm{PPNM}}$. Moreover, abundances are subject to the ANC and ASC in (1).

The above constrained optimization problem is compact, which offers the possibility of integrating various regularization terms into the method. Consequently, considering that the spatial prior favors the critical spatial feature of hyperspectral data and can make the unmixing results approach to the ground truth, we provided a framework to combine spatial regularizer into (23). In this article, the powerful TV regularization term is used as a paradigm, which has already shown its superiority in some hyperspectral linear unmixing algorithms.

The introduction of the TV regularization could make the unmixing process run on the local neighboring pixels rather than the individual pixel. But the complexity of this nonlinear unmixing problem keeps acceptable and can be easily solved. It is worth noting that the TV regularization term is only imposed on the abundances. In this sense, similar to (23), the optimization subproblem with respect to the abundance matrix $\boldsymbol{A} = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_N]$ is actually a linear unmixing problem regularized by the TV regularization term, which has already been well addressed in [40]. Additionally, we should point out that the spatial regularization can be replaced by other state-of-the-art ones, which provides good generalization. The constrained optimization problem combined with the TV regularization term is constructed as

$$\begin{aligned}\boldsymbol{A}^*, \boldsymbol{B}^* = \underset{\boldsymbol{A}, \boldsymbol{B}}{\arg\min}\ &\sum_{j=1}^{N}\left(\frac{1}{2}\|\boldsymbol{e}_j\|^2 + \frac{\gamma}{2}\boldsymbol{\beta}_j^{\mathrm{T}}\boldsymbol{K}\boldsymbol{\beta}_j\right) + \mu J_{\mathrm{sp}}(\boldsymbol{A}) \\ &\text{s.t. } \boldsymbol{A}^{\mathrm{T}}\mathbf{1}_R = \mathbf{1}_N, \boldsymbol{A} \geq 0, \boldsymbol{x}_j = \boldsymbol{M}\boldsymbol{a}_j + \boldsymbol{K}\boldsymbol{\beta}_j + \boldsymbol{e}_j \\ &(j = 1, \ldots, N)\end{aligned} \tag{24}$$

where $\boldsymbol{B} = [\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_N]$, and the regularization function can be written as

$$J_{\mathrm{sp}}(\boldsymbol{A}) = \sum_{n=1}^{N}\sum_{m\in\mathcal{N}(n)}\|\boldsymbol{a}_n - \boldsymbol{a}_m\|_1 \tag{25}$$

where $\mathcal{N}(n)$ denotes the set of neighbors of the pixel n, and μ is a positive parameter to balance the regularization. In this article, we choose the top, bottom, left, and right four points as neighbors of each pixel. To facilitate subsequent calculation,

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a constant matrix $\boldsymbol{H}$ can be derived, and (25) can be rewritten as follows:

$$J_{\mathrm{sp}}(\boldsymbol{A}) = \|\boldsymbol{AH}\|_{1,1} \tag{26}$$

where $\boldsymbol{H}$ is an $N \times 4N$ matrix and $\|\cdot\|_{1,1}$ is the sum of the L1-norms of the columns of a matrix.

Introducing two auxiliary variables $\boldsymbol{U}$ and $\boldsymbol{V}$, we rewrite (24) regularized by (26) as follows:

$$\begin{aligned}\boldsymbol{A}^*, \boldsymbol{B}^* = \arg\min_{\boldsymbol{A},\boldsymbol{B}} &\sum_{j=1}^{N}\left(\frac{1}{2}\|\boldsymbol{e}_j\|^2 + \frac{\gamma}{2}\boldsymbol{\beta}_j^{\mathrm{T}}\boldsymbol{K}\boldsymbol{\beta}_j\right) + \mu\|\boldsymbol{U}\|_{1,1} \\ &\text{s.t. } \boldsymbol{A}^{\mathrm{T}}\boldsymbol{1}_R = \boldsymbol{1}_N, \boldsymbol{A} \geq \boldsymbol{0}, \boldsymbol{V} = \boldsymbol{A}, \boldsymbol{U} = \boldsymbol{VH} \\ &\boldsymbol{x}_j = \boldsymbol{M}\boldsymbol{a}_j + \boldsymbol{K}\boldsymbol{\beta}_j + \boldsymbol{e}_j \\ &(j = 1, \ldots, N).\end{aligned} \tag{27}$$

With the use of the Lagrange multipliers, the augmented Lagrangian function [42] has the following form:

$$\begin{aligned}&L(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{U}, \boldsymbol{V}, \boldsymbol{D}_1, \boldsymbol{D}_2) \\ &= \sum_{i=1}^{N}\left(\frac{1}{2}\|\boldsymbol{x}_j - \boldsymbol{M}\boldsymbol{a}_j - \boldsymbol{K}\boldsymbol{\beta}_j\|^2 + \frac{\gamma}{2}\boldsymbol{\beta}_j^{\mathrm{T}}\boldsymbol{K}\boldsymbol{\beta}_j\right) \\ &\quad + \mu\|\boldsymbol{U}\|_{1,1} + \frac{\xi}{2}\|\boldsymbol{A} - \boldsymbol{V} - \boldsymbol{D}_1\|_{\mathrm{F}}^2 + \frac{\xi}{2}\|\boldsymbol{U} - \boldsymbol{VH} - \boldsymbol{D}_2\|_{\mathrm{F}}^2 \\ &\quad \text{s.t. } \boldsymbol{A}^{\mathrm{T}}\boldsymbol{1}_R = \boldsymbol{1}_N, \quad \boldsymbol{A} \geq \boldsymbol{0}\end{aligned} \tag{28}$$

where $\|\cdot\|_{\mathrm{F}}^2$ denotes the Frobenius norm and ξ is a small positive parameter. This problem is solved by the ADMM, and converted into a group of simple subproblems to search better solutions with respect to $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{U}, \boldsymbol{V}, \boldsymbol{D}_1$, and $\boldsymbol{D}_2$. Specifically, there are five steps described below to be performed.

1) Step1: Update $\boldsymbol{B}$: Since there are no global constraints on the nonlinear parameter $\boldsymbol{B}$, we can update it based on each pixel. Thus, after removing irrelevant terms, the corresponding subproblem is given as follows:

$$\begin{aligned}\boldsymbol{\beta}_j^{(k+1)} = \arg\min_{\boldsymbol{\beta}_j} &\frac{1}{2}\left\|\boldsymbol{y}_j - \boldsymbol{M}\boldsymbol{a}_j^{(k)} - \boldsymbol{K}\boldsymbol{\beta}_j\right\|^2 + \frac{\gamma}{2}\boldsymbol{\beta}_j^{\mathrm{T}}\boldsymbol{K}\boldsymbol{\beta}_j \\ &(j = 1, \ldots, N).\end{aligned} \tag{29}$$

This subproblem is strictly convex, and it is easy to get the following solution:

$$\boldsymbol{\beta}_j^{(k+1)} = (\boldsymbol{K} + \gamma\boldsymbol{I})^{-1}\left(\boldsymbol{x}_j - \boldsymbol{M}\boldsymbol{a}_j^{(k)}\right). \tag{30}$$

After obtaining the nonlinear parameters, all terms related to the nonlinearity can be fixed, and the next four steps can be regarded as the steps of linear unmixing with spatial priors.

2) Step2: Update $\boldsymbol{A}$: By introducing $\boldsymbol{V}$, the optimization of $\boldsymbol{A}$ of each pixel in (28) is independent of each other. For each pixel, the following expression can be given:

$$\begin{aligned}\boldsymbol{a}_j^{(k+1)} = \arg\min_{\boldsymbol{a}_j} &\frac{1}{2}\left\|\boldsymbol{x}_j - \boldsymbol{M}\boldsymbol{a}_j - \boldsymbol{K}\boldsymbol{\beta}_j^{(k)}\right\|^2 \\ &+ \frac{\xi}{2}\left\|\boldsymbol{a}_j - \boldsymbol{v}_j^{(k)} - \boldsymbol{d}_{1,j}^{(k)}\right\|^2 \\ &\text{s.t. } 1^{\mathrm{T}}\boldsymbol{a}_j = 1, \quad \boldsymbol{a}_j \geq \boldsymbol{0}\end{aligned} \tag{31}$$

where $\boldsymbol{v}_j$ is the column j of $\boldsymbol{V}$, and $\boldsymbol{d}_{1,j}$ is the column j of $\boldsymbol{D}_1$. Since it is strictly a quadratic programming problem, we can solve it using the quadratic programming toolbox in MATLAB.

3) Step3: Update $\boldsymbol{V}$: After fixing the other variables, the optimization problem is reduced to

$$\boldsymbol{V}^{(k+1)} = \arg\min_{\boldsymbol{V}}\left\|\boldsymbol{A}^{(k)} - \boldsymbol{V} - \boldsymbol{D}_1^{(k)}\right\|_{\mathrm{F}}^2 + \left\|\boldsymbol{U}^{(k)} - \boldsymbol{VH} - \boldsymbol{D}_2^{(k)}\right\|_{\mathrm{F}}^2 \tag{32}$$

which is a convex problem. The closed-form solution is

$$\boldsymbol{V} = \left(\boldsymbol{A}^{(k)} - \boldsymbol{D}_1^{(k)} + \left(\boldsymbol{U}^{(k)} - \boldsymbol{D}_2^{(k)}\right)\boldsymbol{H}^{\mathrm{T}}\right)\left(\boldsymbol{I} + \boldsymbol{H}\boldsymbol{H}^{\mathrm{T}}\right)^{-1}. \tag{33}$$

4) Step4: Update $\boldsymbol{U}$: Similarly, the subproblem with respect to $\boldsymbol{U}$ can be provided in the following equation:

$$\boldsymbol{U}^{(k+1)} = \arg\min_{\boldsymbol{U}} \mu\|\boldsymbol{U}\|_{1,1} + \frac{\xi}{2}\left\|\boldsymbol{U} - \boldsymbol{V}^{(k)}\boldsymbol{H} - \boldsymbol{D}_2^{(k)}\right\|_{\mathrm{F}}^2. \tag{34}$$

In spite of the direct use of $\|\cdot\|_{1,1}$, the soft threshold operator [46] $\mathrm{Thresh}(\cdot)$ is adopted

$$\boldsymbol{U} = \mathrm{Thresh}\left(\boldsymbol{V}^{(k)}\boldsymbol{H} + \boldsymbol{D}_2^{(k)}, \frac{\mu}{\xi}\right). \tag{35}$$

5) Step5: Update $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$: In the last step, the Lagrange multipliers $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are updated according to the following equations:

$$\boldsymbol{D}_1^{(k+1)} = \boldsymbol{D}_1^{(k)} + \left(\boldsymbol{V}^{(k)} - \boldsymbol{A}^{(k)}\right) \tag{36}$$

$$\boldsymbol{D}_2^{(k+1)} = \boldsymbol{D}_2^{(k)} + \left(\boldsymbol{V}^{(k)}\boldsymbol{H} - \boldsymbol{U}^{(k)}\right). \tag{37}$$

The above alternating iterations run until the convergence conditions are met. In this article, both the maximum number of iterations $T_{\max}$ and $\|\boldsymbol{A}^{(k)} - \boldsymbol{A}^{(k-1)}\|_{\mathrm{F}} < \eta$ are set as the stop criteria. $\boldsymbol{A}$ and $\boldsymbol{V}$ are both initialized by the abundances provided by FCLS, then $\boldsymbol{U} = \boldsymbol{AH}$ can be obtained. $\boldsymbol{B}$, $\boldsymbol{D}_1$, $\boldsymbol{D}_2$ are initialized by zeros. The proposed algorithm named KBU-TV is summarized in Algorithm 1. As for the computational complexity, the update of $\boldsymbol{V}$ in (33) and the update of $\boldsymbol{U}$ in (35) take the highest computational burden. At each iteration, the complexities of (33) and (35) are $\mathcal{O}(RN^2)$, which are strictly constrained by the size of image. Other steps have less computational complexities $\mathcal{O}(L^2N)$ or $\mathcal{O}(RN)$. Moreover, the execution time of KBU-TV will be discussed in the real data sets experiment.

IV. Experimental Results

In this section, the synthetic data, the physical-based simulated data of virtual vegetated areas, and real HSIs are prepared for the experiments, and the proposed methods are evaluated by comparing them with other state-of-the-art algorithms. Through these experiments, the performance of the proposed methods on abundance estimation could be tested, and their ability to fit on the nonlinear parts could also be validated.

A. Compared Algorithms and Performance Metrics

In our experiments, we have compared seven selected algorithms. The first one is FCLS [9], which is widely used for linear unmixing. The second one is the GBM-based gradient descent algorithm (GBMGDA) [18], which is designed based

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on the GBM. Then the gradient descent algorithm for the PPNM (PPNMGDA) [19] is also set as a compared algorithm in our experiments. Besides, two kernel-based algorithms are included as well, i.e., kernel-based FCLS (KFCLS) [33] and K-Hype [32]. For KFCLS, the Gaussian kernel is used; For K-Hype, both the Gaussian kernel and polynomial kernel are considered, and the corresponding algorithms are denoted as K-Hype(G) and K-Hype(P), respectively. Finally, two tensor factorization-based algorithms (low-rank based nonnegative tensor factorization (LR-NTF) [22] and plug-and-play framework-based nonnegative tensor factorization (PnP-NTF) [23]) are set up as comparison methods, which can exploit the spatial correlation of abundance maps in the nonlinear unmixing.

When the spatial priors are not considered, we set $\mu = 0$ and denote the Algorithm 1 on $\boldsymbol{K}_{\text{GBM}}$ and $\boldsymbol{K}_{\text{PPNM}}$ as KBU(GBM) and KBU(PPNM), respectively. Also, the polynomial kernel in the following equation is used in KBU for the purpose of validating the effectiveness of the proposed $\boldsymbol{K}_{\text{GBM}}$ and $\boldsymbol{K}_{\text{PPNM}}$, and the corresponding algorithm is denoted as KBU(P):

$$\begin{aligned} \boldsymbol{K}_{\text{polynoimal}} &= (\boldsymbol{1} + \boldsymbol{M}\boldsymbol{M}^{\text{T}})^2 \\ \left[\boldsymbol{K}_{\text{polynomial}}\right]_{i,j} &= \kappa(\tilde{\boldsymbol{m}}_i, \tilde{\boldsymbol{m}}_j) = \left(1 + \tilde{\boldsymbol{m}}_i \tilde{\boldsymbol{m}}_j^{\text{T}}\right)^2. \end{aligned} \tag{38}$$

The algorithms incorporating the TV regularization term into KBU are marked as KBU(GBM)-TV and KBU(PPNM)-TV, respectively. All the algorithms are executed on a computer with a 3.79-GHz AMD Ryzen 9 3900X CPU and 32 GB of memory.

We use two performance metrics to quantitatively evaluate the performance of each algorithm on the data sets. The first metric is the root-mean-square error (RMSE) of abundances, which is defined in (39), and it is used for evaluating the accuracy of abundance estimation. The second one is the reconstruction error of the nonlinear parts (REN), aiming to test the fitting accuracy of the nonlinear parts, and it is defined in (40)

$$\text{RMSE} = \sqrt{\frac{1}{NR}\sum_{j=1}^{N}\left\|\boldsymbol{a}_j - \boldsymbol{a}_j^*\right\|_2^2} \tag{39}$$

$$\text{REN} = \sqrt{\frac{1}{NL}\sum_{j=1}^{N}\left\|\boldsymbol{f}_j - \boldsymbol{f}_j^*\right\|_2^2}. \tag{40}$$

In (39), $\boldsymbol{a}_j$ and $\boldsymbol{a}_j^*$ denote true abundances and estimated abundances, respectively. $\boldsymbol{f}_j$ and $\boldsymbol{f}_j^*$ in (40) represent the real nonlinear part in a pixel and the estimated nonlinear part, respectively.

B. Experiments With Synthetic Data Sets

1) Data Sets Generation: To give a detailed analysis and validate the effectiveness of the proposed algorithms, we produce a synthetic data set, called data cube 0 (DC0) with 50×50 pixels. The data set consists of three spectral signatures selected from the USGS Spectral Library.[1] As shown in Fig. 1, they are all available in 224 bands and cover a spectral range of wavelengths from 380 to 2500 nm. Then, the Gaussian fields method [47] was used here to generate spatially correlated maps for DC0. Moreover, two BMMs GBM (GBM includes the case of FM) and PPNM are used as the basic models for the generation of pixels in the data sets. In terms of the GBM, the nonlinear parameter $\gamma_{i,j,k}$ in (3) is randomly selected from set $[0, 1]$. For the PPNM, the nonlinear parameter b_j in (4) is randomly selected from set $[-0.3, 0.3]$ as reported in [19]. Finally, Gaussian noises with signal-to-noise ratios (SNRs) of 20 and 30 dB are added to all the data sets, respectively.

[1]Available at http://speclab.cr.usgs.gov/spectral-lib.html.

Algorithm 1 KBU-TV

Input: hyperspectral data matrix $\boldsymbol{X} \in \mathbb{R}^{L \times N}$, endmember matrix $\boldsymbol{M} \in \mathbb{R}^{L \times R}$, the regularization parameters γ, μ, and the penalty parameter ξ.

Initialize $(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{U}, \boldsymbol{V}, \boldsymbol{D}_1, \boldsymbol{D}_2)$, and set $\boldsymbol{K}$ according to the case. Set $k = 0$
While stop conditions are not met, **do**
a) Update $\boldsymbol{B}$ with (30).
b) Update $\boldsymbol{A}$ with (31).
c) Update $\boldsymbol{V}$ with (33).
d) Update $\boldsymbol{U}$ with (35).
e) Update $\boldsymbol{D}_1, \boldsymbol{D}_2$ with (36) and (37).
f) $k = k + 1$.
End
Output: The estimated abundance matrix $\boldsymbol{A}^* = [\boldsymbol{a}_1^*, \ldots, \boldsymbol{a}_N^*]$ and the nonlinear. parameter matrix $\boldsymbol{B}^* = [\boldsymbol{\beta}_1^*, \ldots, \boldsymbol{\beta}_N^*]$

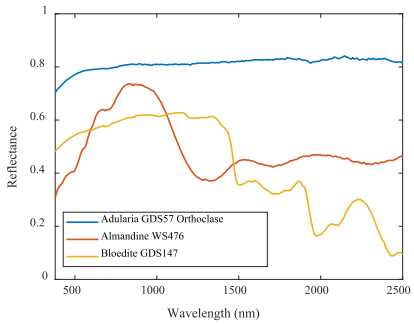


Fig. 1. Endmember spectra selected from the USGS Spectral Library.

2) Parameter Settings of the Algorithms: To guarantee the fairness of the experimental results, the parameters of each algorithm in the experiments are tuned according to the corresponding works. For FCLS, the parameter υ to control ASC is set as 10^{-3} [9], and the abundances estimated by FCLS are set as the initialization of other nonlinear unmixing algorithms. The KFCLS has two parameters, the bandwidth σ of the Gaussian kernel, and the parameter υ to control ASC. They are set as 0.01 and 1, respectively [33]. For the GBM, the maximum number of iterations $T_{\max}$ is 1000 and the tolerance parameter is 10^{-6}[18]. Similarly, for the PPNM, the maximum number of iterations $T_{\max}$ is set as 3000 and the tolerance parameter is set as 10^{-8} [19]. In terms of K-Hype, two kernels are used in the experiments, i.e., Gaussian kernel and polynomial kernel. The bandwidth σ of the Gaussian kernel is set as 3, and the polynomial kernel takes the form of (11) and set $d = 2, c = 1$ [32]. And the tradeoff parameter

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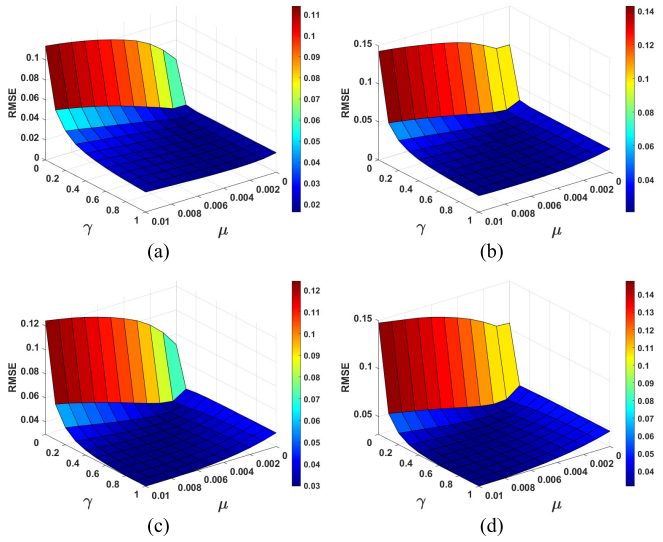


Fig. 2. RMSE variations of the KBU(GBM)-TV under different combinations of tradeoff parameters γ and μ: (a) GBM-30 dB; (b) GBM-20 dB; (c) PPNM-30 dB; and (d) PPNM-20 dB.

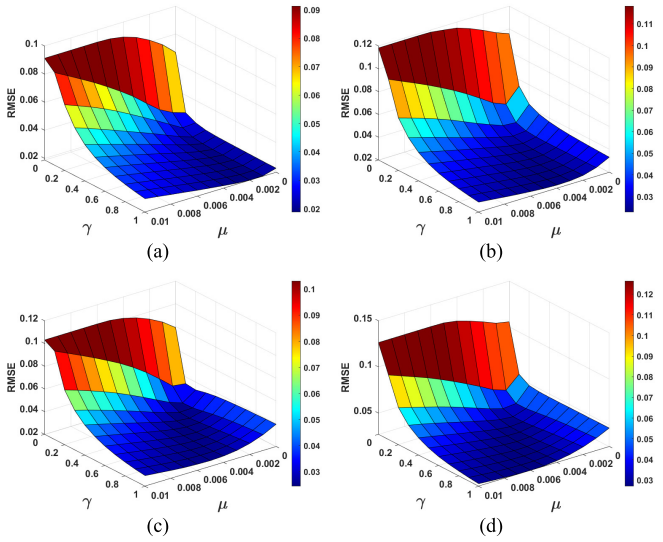


Fig. 3. RMSE variations of the KBU(PPNM)-TV under different combinations of tradeoff parameters γ and μ: (a) GBM-30 dB; (b) GBM-20 dB; (c) PPNM-30 dB; and (d) PPNM-20 dB.

μ in K-Hype is set as 5×10^{-3} [32]. For both the tensor factorization-based algorithms, the parameters of LR-NTF [22] are set as $\lambda_1 = 0.1, \lambda_2 = 0.07, \mu = 8 \times 10^{-3}, T_{\max} = 1000$, and the parameters of PnP-NTF [23] are set as $\mu = 8 \times 10^{-3}$, $T_{\max} = 1000$, following the parameter setting in [22] and [23], respectively.

For the proposed algorithms KBU(GBM), KBU(PPNM), KBU(GBM)-TV, KBU(PPNM)-TV, the maximum number of iterations $T_{\max}$ is set as 500, the tolerance η is specified as 10^{-5}, and the penalty parameter ξ that controls the convergence speed of ADMM is set as 0.001. Besides, a group of experiments on DC0 are conducted to select the tradeoff parameter γ in the range of $(0, 1)$ and the regularization parameter μ in the range of $(0, 0.01)$. The RMSE results are displayed in Figs. 2 and 3. Without regard to the spatial priors, that is, set $\mu = 0$, it can be observed that $\gamma = 0.2$ for KBU(GBM) and $\gamma = 0.5$ for KBU(PPNM) could lead to better unmixing results overall. When taking the TV regularization term into account, we can make the observation that $\gamma = 0.2$, $\mu = 0.003$ is more appropriate for KBU(GBM)-TV and $\gamma = 0.5, \mu = 0.003$ is more appropriate for KBU(PPNM)-TV in the later experiments.

3) Unmixing Results and Discussion: The RMSEs and RENs of unmixing results on DC0 using the proposed methods and the comparison methods are summarized in Table II, and the best results are marked with bold font.

a) The effectiveness of the designed kernels: First, we focused on the KBU with different kernels, i.e., KBU(GBM), KBU(PPNM), and KBU(P). We could observe that the KBU(GBM) could always get the best RMSE and RE results on all the GBM-based data sets, and KBU(PPNM) could always play the best on all the PPNM-based data sets, which is inferred that the Gram matrix $\boldsymbol{K}_{\text{GBM}}$ is more suitable for the GBM and Gram matrix $\boldsymbol{K}_{\text{PPNM}}$ is more suitable for the PPNM, because the bilinear components in $\boldsymbol{K}_{\text{GBM}}$ correspond to the nonlinear components of GBM, and the same for $\boldsymbol{K}_{\text{PPNM}}$. $\boldsymbol{K}_{\text{polynomial}}$ is the traditional polynomial kernel, and the KBU(P) performs worse than KBU(GBM) and KBU(PPNM) in all data sets. Obviously, the RENs of KBU(P) cannot keep up with the other two, which indicates it is not able to achieve an effective fitting for the nonlinear parts of the pixel; that is, it means that the zero-order and first-order components in $\boldsymbol{K}_{\text{polynomial}}$ indeed has a negative impact.

b) Comparison with other unmixing algorithms: Quite apart from spatial information, KBU(GBM) and KBU(PPNM) outperformed the competing algorithms for all the cases on RMSEs and RENs. It is worth noting that K-Hype with two kernels cannot get the desired unmixing results, and the REN results of K-Hype are worse than other nonlinear unmixing algorithms, indicating that the K-Hype has severe overfitting for the nonlinear parts of the pixels and thus degrades the abundance estimation. Besides, compared with the traditional GDA algorithms for GBM and PPNM, it can be observed from the results that KBU(GBM) and KBU(PPNM) seem to converge to more optimal solutions easily. The abundance maps of DC0 for the case of GBM with an SNR of 20 dB provided by all the compared methods are displayed in Fig. 4 for illustrative purposes. It can be seen that the abundance maps from KBU(GBM) better approximates the ground truth, which is in accordance with the RMSE results as well.

c) The effectiveness of the spatial regularization: When taking the spatial information into account, it can be seen from Table II that KBU-TV achieves a considerable improvement in abundance estimation on the basis of KBU because the valuable spatial prior information can be exploited. Compared with both the tensor factorization-based methods (LR-NTF and PnP-NTF), the results of KBU-TV still show the better abundance estimation and nonlinear reconstruction. In Fig. 4, the abundance maps of KBU(GBM)-TV and KBU(PPNM)-TV are smoother and very similar to the ground truth. Due to the expansibility of the KBU-based method, other state-of-the-art spatial regularizations can also be taken into account to obtain better unmixing accuracy.

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TABLE II
RMSE AND REN RESULTS FOR DC0

SNR	Method	GBM		PPNM	
		RMSE	REN	RMSE	REN
30dB	FCLS [9]	0.0760±0.0000	0.0387±0.0000	0.0893±0.0000	0.0478±0.0000
	KFCLS [33]	0.0846±0.0000	-	0.1099±0.0000	-
	GBM-GDA [18]	0.0468±0.0000	0.0207±0.0000	0.0815±0.0000	0.0414±0.0000
	PPNM-GDA [19]	0.0270±0.0000	0.0089±0.0000	0.0489±0.0000	0.008±0.0000
	K-Hype(P) [32]	0.0556±0.0002	0.0245±0.0001	0.0721±0.0003	0.0321±0.0001
	K-Hype(G) [32]	0.0386±0.0002	0.0180±0.0001	0.0568±0.0002	0.0264±0.0001
	LR-NTF [22]	0.0185±0.0000	0.0124±0.0000	0.0505±0.0000	0.0479±0.0000
	PnP-NTF [23]	0.0189±0.0000	0.0140±0.0000	0.0503±0.0000	0.0476±0.0000
	KBU (P) (ours)	0.0283±0.0000	0.0125±0.0000	0.0449±0.0000	0.0206±0.0000
	KBU(GBM) (ours)	0.0187±0.0000	0.0051±0.0000	0.0393±0.0000	0.0166±0.0000
	KBU(PPNM) (ours)	0.0204±0.0000	0.0054±0.0000	0.0383±0.0000	0.0155±0.0000
	KBU(GBM)-TV (ours)	**0.0166±0.0000**	**0.0048±0.0000**	0.0307±0.0000	0.0132±0.0000
	KBU(PPNM)-TV (ours)	0.0190±0.0000	0.0069±0.0000	**0.0277±0.0000**	**0.0116±0.0000**
20dB	FCLS [9]	0.0779±0.0001	0.0390±0.0000	0.0912±0.0001	0.0484±0.0000
	KFCLS [33]	0.0879±0.0001	-	0.1142±0.0001	-
	GBM-GDA [18]	0.0501±0.0001	0.0210±0.0000	0.0833±0.0001	0.0413±0.0001
	PPNM-GDA [19]	0.0337±0.0001	0.0103±0.0000	0.0541±0.0001	0.0106±0.0000
	K-Hype(P) [32]	0.0806±0.0005	0.0378±0.0002	0.0908±0.0006	0.0423±0.0001
	K-Hype(G) [32]	0.0669±0.0004	0.0320±0.0001	0.0771±0.0005	0.0367±0.0001
	LR-NTF [22]	0.0275±0.0001	0.0132±0.0000	0.0549±0.0002	0.0474±0.0001
	PnP-NTF [23]	0.0272±0.0001	0.0137±0.0000	0.0548±0.0003	0.0474±0.0001
	KBU(P) (ours)	0.0382±0.0001	0.0147±0.0001	0.0519±0.0003	0.0220±0.0001
	KBU(GBM) (ours)	0.0305±0.0000	0.0090±0.0000	0.0472±0.0002	0.0175±0.0001
	KBU(PPNM) (ours)	0.0329±0.0001	0.0098±0.0000	0.0467±0.0002	0.0171±0.0001
	KBU(GBM)-TV (ours)	**0.0225±0.0000**	**0.0078±0.0000**	0.0326±0.0000	0.0141±0.0000
	KBU(PPNM)-TV (ours)	0.0247±0.0000	0.0082±0.0000	**0.0309±0.0000**	**0.0129±0.0000**

TABLE III
UNMIXING RESULTS OF THE ALGORITHMS FOR FOREST DATA SETS

Method	Frst_b20p80		Frst_b40p60		Frst_b60p40		Frst_b80p20	
	RMSE	REN	RMSE	REN	RMSE	REN	RMSE	REN
FCLS [9]	0.0462	0.0084	0.0608	0.0116	0.0605	0.0129	0.0568	0.0082
KFCLS [33]	0.0829	-	0.0755	-	0.0608	-	0.0570	-
GBM-GDA [18]	0.0457	0.0081	0.0610	0.0110	0.0606	0.0131	0.0558	0.0074
PPNM-GDA [19]	0.0449	0.0069	0.0568	0.0095	0.0546	0.0101	0.0554	0.0074
K-Hype(P) [32]	0.0865	0.0112	0.0763	0.0133	0.0701	0.0102	0.0900	0.0097
K-Hype(G) [32]	0.0775	0.0106	0.0652	0.0109	0.0532	0.0098	0.0727	0.0094
KBU(GBM) (ours)	0.0419	0.0057	0.0526	0.0085	0.0555	0.0096	0.0534	0.0076
KBU(PPNM) (ours)	**0.0366**	**0.0052**	**0.0465**	**0.0071**	**0.0490**	**0.0082**	**0.0474**	**0.0071**

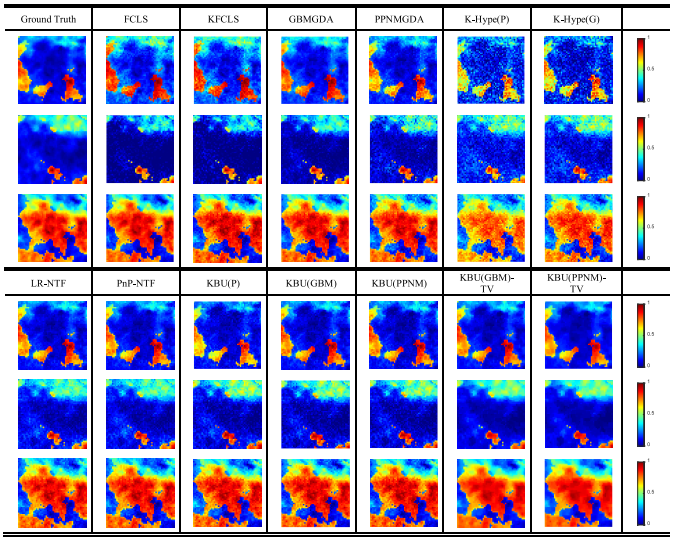


Fig. 4. Abundance maps estimated by all algorithms for the DC0.

C. Experiments With Physical-Based Simulated Data Sets

Experiments on synthetic data sets can test the effectiveness of the KBU-based methods and the ability to fit nonlinearities. Nevertheless, the spectra of ground objects cannot be mixed precisely in the way of GBM and PPNM. Therefore, the physical-based simulated data sets of forests in [48] are used in this section to further demonstrate the effectiveness of our proposed methods. These data sets are well calibrated by *in situ* data that the detailed light transmission paths are known, and multiple scatterings are well simulated.

1) *Data Description:* There are four data sets in total, each consisting of two kinds of trees, i.e., beech trees and poplar trees, and a small amount of soil. As shown in Fig. 5, the proportions of the two trees in each data set are different, namely, beech tree: 20%, poplar tree: 80%; beech tree: 40%, poplar tree: 60%; beech tree: 60%, poplar tree: 40%; and beech tree: 80%, poplar tree: 20%. Then each data set is named as Frst_b20p80, Frst_b40p60, Frst_b60p40, and Frst_b80p20, respectively. As shown in Fig. 6, spectral variability is present in these data sets, in which each pixel will be unmixed with different endmember matrices. Besides, since these data sets show weak spatial relationships, the KBU-TV, LR-NTF and PnP-NTF are not performed in this experiment.
2) *Experimental Results and Analysis:* the experiments are performed on the eight algorithms introduced in Section IV-B. The RMSEs results are displayed in Table III. It can be seen that KBU(PPNM) always

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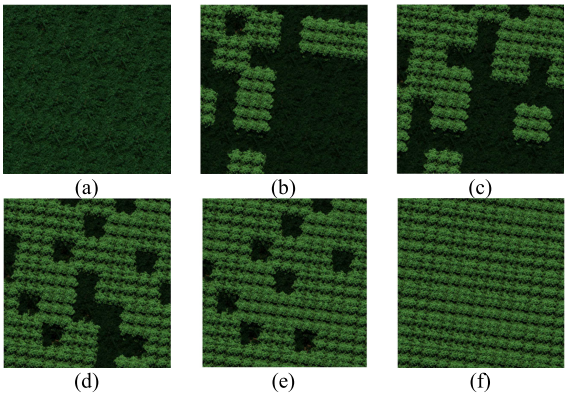


Fig. 5. Forest data sets: (a) 100% beech tree–0% poplar tree; (b) 80% beech tree–20% poplar tree; (c) 60% beech tree–40% poplar tree; (d) 40% beech tree–60% poplar tree; (e) 20% beech tree–80% poplar tree; and (f) 0% beech tree–1000% poplar tree.

obtained the best abundance estimation. And it is worth noting that the RMSEs of KBU(GBM) are not good as those of KBU(PPNM). It is inferred that the bilinear interactions from the same endmembers are strong in the data sets because the data sets mainly consist of two trees. Indirectly, this point can also be confirmed by the comparison of the two methods GBMGDA and PPNMGDA. The K-Hype with Gaussian and polynomial kernels get worse results than most of compared algorithms, which may result from the over-fitting of the nonlinear parts as analyzed in Section IV-B.

D. Experiments on Real HSIs

In this section, we use two HSIs to verify the effectiveness of our algorithms and compare them with other algorithms. Since the compared algorithms are supervised, the endmembers in the images are extracted using the VCA algorithm. To avoid the randomness associated with VCA, each algorithm with VCA was performed independently for ten times. The parameters are set to be consistent with the experiments on the synthetic data sets. Since the abundances of the real HSIs are unknown, RMSE and REN turn to be unusable here. Hence, the abundance maps and nonlinear intensity maps obtained by the algorithms will be compared and analyzed. Moreover, the RE in (41) and the average spectral angle mapping (aSAM) in (42) are used as a reference

$$\mathrm{RE} = \sqrt{\frac{1}{NL}\sum_{j=1}^{N}\left\|\boldsymbol{x}_j - \boldsymbol{x}_j^*\right\|_2^2} \tag{41}$$

$$\mathrm{aSAM} = \frac{1}{N}\sum_{j=1}^{N}\arccos\left(\frac{\boldsymbol{x}_j^{\mathrm{T}}\boldsymbol{x}_j^*}{\left\|\boldsymbol{x}_j\right\|\left\|\boldsymbol{x}_j^*\right\|}\right) \tag{42}$$

where $\boldsymbol{x}_j \in \mathbb{R}^{L\times 1}(j = 1, \ldots, N)$ denotes a pixel.

1) Data Sets Description: The first image is the Jasper Ridge data sets, which was acquired by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). The ground sample distance (GSD) is 20 m and each pixel contains 224 electromagnetic bands ranging from 380 to 2500 nm [49]. Due to the dense water vapor and atmospheric effects, the spectral

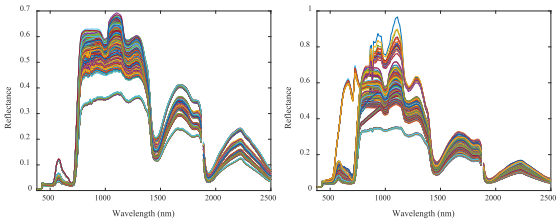


Fig. 6. Endmember spectra for each pixel in the forest data sets for: (a) beech tree and (b) poplar tree.

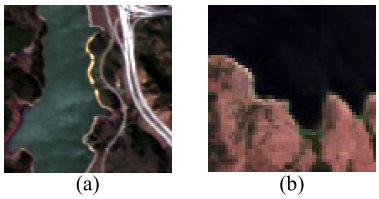


Fig. 7. Subregions of two real HSIs: (a) Jasper Ridge and (b) Moffett.

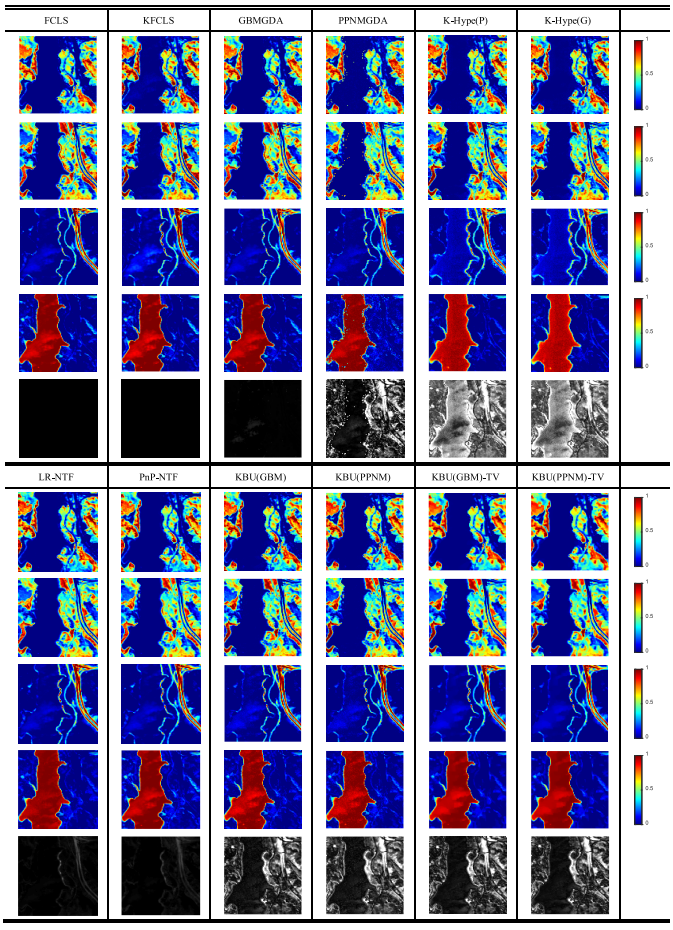


Fig. 8. Abundance maps and nonlinear intensity maps of Jasper Ridge data estimated using different algorithms. For each row: vegetation, soil, road, river, and nonlinear intensity.

bands 1–3, 108–112, 154–166, and 220–224 are removed, remaining 198 bands. As shown in Fig. 7(a), a subimage of 100×100 pixels is selected from the data set for the experiment, and it is known to have four predominant endmembers, i.e., vegetation, road, ground, and water [49].

The second image is the Moffett field HSI, which was captured via the AVIRIS sensor by the Jet Propulsion Laboratory (JPL), Pasadena, CA, USA, in 1997 [50]. This

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TABLE IV
UNMIXING RESULTS OF THE ALGORITHMS FOR REAL HYPERSPECTRAL DATA SETS

Method	Jasper Ridge			Moffett	
	RMSE	RE	aSAM	RE	aSAM
FCLS [9]	0.1478±0.0132	0.0313±0.0060	0.0903±0.0166	0.0065±0.0001	0.1411±0.0015
KFCLS [33]	0.1547±0.0128	0.0167±0.0032	0.0794±0.0070	0.0066±0.0003	0.1347±0.0016
GBM-GDA [18]	0.1477±0.0132	0.0306±0.0061	0.0900±0.0167	0.0064±0.0001	0.1411±0.0015
PPNM-GDA [19]	0.1421±0.0151	0.0175±0.0054	0.0924±0.0179	0.0061±0.0001	0.1450±0.0022
K-Hype(P) [32]	0.1411±0.0053	0.0147±0.0004	0.0765±0.0016	**0.0055±0.0000**	0.1286±0.0000
K-Hype(G) [32]	0.1413±0.0059	**0.0144±0.0003**	**0.0756±0.0012**	**0.0055±0.0000**	0.1287±0.0000
LR-NTF [22]	0.1418±0.0072	0.0171±0.0051	0.0831±0.0139	0.0064±0.0001	0.1406±0.0016
PnP-NTF [23]	0.1417±0.0134	0.0180±0.0058	0.0848±0.0163	0.0064±0.0001	0.1409±0.0015
KBU(GBM) (ours)	0.1375±0.0056	0.0161±0.0009	0.0786±0.0051	0.0063±0.0001	0.1323±0.0017
KBU(PPNM) (ours)	0.1352±0.0067	0.0153±0.0005	0.0760±0.0027	0.0061±0.0000	**0.1266±0.0011**
KBU(GBM)-TV (ours)	0.1340±0.0059	0.0163±0.0009	0.0790±0.0051	0.0064±0.0001	0.1375±0.0015
KBU(PPNM)-TV (ours)	**0.1304±0.0069**	0.0157±0.0005	0.0765±0.0027	0.0063±0.0000	0.1324±0.0007

data set has 224 bands corresponding to the wavelengths from 400 to 2500 nm. After removing the water-vapor absorption and low SNR bands, 185 bands (5–104, 116–149, and 171–221) remained. In this experiment, a subimage containing 50×50 pixels is chosen. Water, soils, and vegetation are the primary ground objects shown in Fig. 7(b).

2) Unmixing Results for Jasper Ridge: Fig. 8 displays the abundance maps and the nonlinearity intensity maps, and Table IV summarizes the unmixing results for Jasper Ridge data set. Fortunately, the ground truth is available for Jasper Ridge, and therefore, the RMSE results can be provided. It can be seen that most algorithms can identify the distribution of ground objects. A detailed analysis of KBU(GBM) and KBU(PPNM) results indicates that they have more soil and road components when compared to other nonlinear unmixing algorithms. Besides, with the help of TV, KBU(GBM)-TV and KBU(PPNM)-TV get smoother abundance maps, making them visually more in line with the ground truth. Furthermore, the nonlinear intensity maps of KBU-based methods show a more reasonable distribution of the nonlinear components that the nonlinear intensity is strong where the adjacency effects may have occurred but is weak where the adjacency effects are less likely to appear. Owing to the available ground truth, the RMSE results can be obtained. And from the results, the KBU-TV methods perform the best, which fully illustrates numerically the superiority of our proposed method for abundance estimation. And the aSAM results of KBU are also excellent compared with other methods. The RE results show that the K-Hype (G) gets the best score, but it may result from overfitting, which can be seen from its nonlinear intensity map with the unreasonable spatial distribution of nonlinear intensity. For example, the nonlinearity in the water portion is supposed to be relatively small, yet the results from K-Hype (G) show relatively large nonlinearity compared with the other methods, which may be due to the fact that K-Hype (G) may have incorrectly fitted the other components as nonlinearities.

3) Unmixing Results for Moffett Field: The abundance maps and nonlinearity intensity maps for Moffett field data set are displayed in Fig. 9, and the corresponding RE results are summarized in Table IV. We can observe that the KBU(PPNM)-TV reflects the best spatial distributions of the ground objects. In particular, the distribution of water is clearly determined and seems more in line with the ground truth. Besides, it can be figured out from the nonlinearity intensity maps in Fig. 9 that the nonlinearity at the junction of soil and water is highlighted

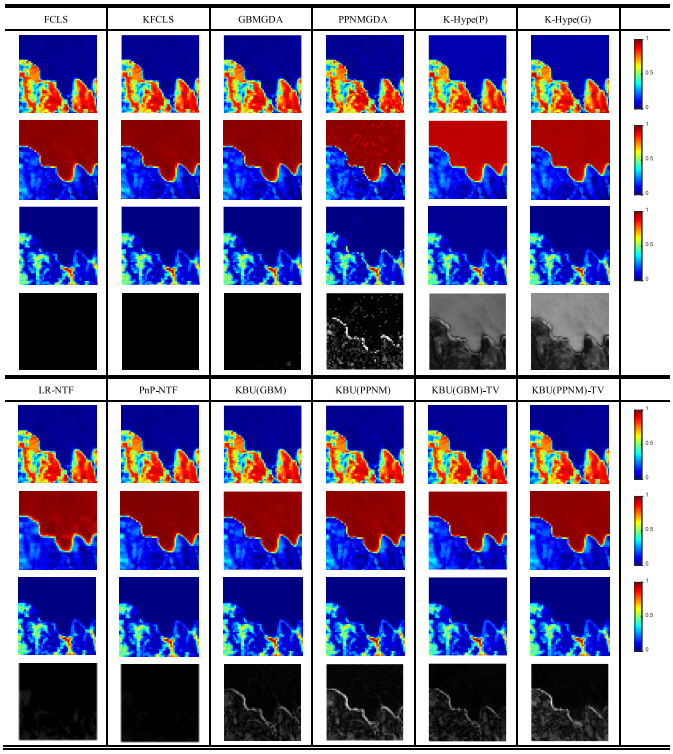


Fig. 9. Abundance maps and nonlinear intensity maps of Moffett data estimated using different algorithms. For each row: soil, water, vegetation, and nonlinear intensity.

TABLE V
EXECUTION TIME (IN S) OF THE ALGORITHMS

Method	Jasper Ridge	Moffett
FCLS [9]	**0.57**	**0.53**
KFCLS [33]	0.60	0.58
GBM-GDA [18]	16.82	2.55
PPNM-GDA [19]	4.010	1.457
K-Hype(P) [32]	30.46	4.31
K-Hype(G) [32]	40.21	4.32
LR-NTF [22]	50.78	7.19
PnP-NTF [23]	632.82	115.73
KBU(GBM) (ours)	11.25	3.69
KBU(PPNM) (ours)	12.59	4.16
KBU(GBM)-TV (ours)	79.29	43.02
KBU(PPNM)-TV (ours)	79.11	42.37

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by KBU-based methods. The K-Hype methods get the lowest RE results, but as analyzed above and displayed in the nonlinearity intensity maps in Fig. 9, it may result from overfitting. However, the KBU(PPNM) obtains the best aSAM result and other KBU-based methods yield the similar good results, indicating that KBU has good pixel reconstruction power.

4) Computational Analysis: The execution time of the algorithms is summarized in Table V. FCLS and KFCLS performed much better in execution time than the other algorithms. As the bilinear unmixing methods, the proposed KBU takes the comparable execution time with GBMGDA and PPNMGDA. Moreover, compared with K-Hype, the proposed KBU is a kernel-based method as well, but KBU is much faster and performs the unmixing efficiently. When considering the spatial regularization, the TV imposes a computational burden on KBU-TV. It can be seen that the execution times of KBU-TV are higher than LR-NTF but lower than PnP-NTF. If this global constraint of TV can be broken, then the algorithm will run significantly faster. Therefore, it is hopeful that it is able to introduce other nonglobal spatial regularizations to improve the KBU. And, owing to the properties of KBU, it is as easy to introduce spatial regularizations as in linear unmixing.

V. Conclusion

In this article, a KBU method was proposed, which achieved a flexible nonparametric representation of the nonlinear part (for bilinear components) of the pixel through kernel transformations. In particular, we focused on the reconstruction of the nonlinearity from mixing mechanism rather than data-driven way, and two appropriate kernels κ_{GBM} and κ_{PPNM} were designed for this method. For the construction of the kernels, we removed the components of the kernels that were not related to the nonlinearity in the pixel, which makes the reconstruction of the nonlinear part more accurate. Then, the corresponding constrained optimization problem was deduced and solved by the ADMM method to estimate the abundances and nonlinear parameters. Due to the conciseness of the proposed method, a framework that incorporates a spatial prior into the optimization problem can be derived. Specifically, the unmixing problem is transformed into a linear unmixing issue, which can greatly reduce its complexity. A KBU-TV method that integrates the TV regularizer was proposed as a paradigm. Experiments on synthetic data sets, physical-based simulated data sets, and real data sets have demonstrated that our methods could achieve an effective fit to the nonlinearities and obtain excellent abundance estimation performance compared with other state-of-the-art algorithms.

Moreover, since the proposed KBU-based algorithms still require the extracted endmembers in advance, our future work will study the unsupervised version of KBU.

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